Anoushkrit Goel

Researcher | **Data Scientist** | Entrepreneur
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Research Interests

Foundation models for **medical** and **3D point-cloud** data (neuroimaging, tractography, white matter analysis); reinforcement and representation learning; multimodal generation and editing of 3D shapes; document AI for CAD and architectural drawings; biomedical signal processing.

Education

Indian Institute of Technology Mandi, M.Tech by Research in *Intelligent Systems* – **9.47/10.0** Feb 2023 – Jun 2025

- Advisors: Prof. Arnav Bhavsar and Prof. Aditya Nigam. Research funded by SERB / DST India.
- Coursework: Deep Learning, Advanced Deep Learning, Biomedical Systems, Mobile Virtual Reality and AI, Research Methodology

Shiv Nadar University, B.Tech in Electronics and Communication Engineering – 7.61/10.0 Jul 2015 – May 2019

- Minor: Design. Bachelor's thesis: hardware for multimodal cardiac signal capture (basis of Tensr.AI).

Delhi Public School Ghaziabad: Class 12th (PCM) 92.4% 2015
Delhi Public School Ghaziabad: Class 10th – 10 CGPA 2013

Positions

Research Engineer 2, *Attentive.AI* – Gurgaon, IN current Jun 2025 – Present

- Deep learning for symbol search, vector-path (.svg) annotation and HVAC-element annotation on complex architectural PDFs.
- Tokenization approach for pretraining on vector paths: +7% class-wise accuracy over state-of-the-art CAD segmentation models.
- Cascaded multi-OCR design for fast, robust text-callout extraction from CAD drawings.

Researcher, *IIT Mandi* – Mandi, IN Feb 2023 – Jun 2025

- Foundation models for medical and 3D point-cloud data using unsupervised, reinforcement and supervised learning; state-of-the-art white matter tract segmentation.
- Built an end-to-end open-source pipeline on 3D Slicer (**SlicerTracto**), being deployed at **PGI Chandigarh**.
- Outperformed Harvard Medical School (TractCloud) and SCIL, Canada (FIESTA); 5 papers at ICPR, ISBI.

Lead Data Scientist, *StatusNeo* – Gurgaon, IN Aug 2022 – Mar 2023

- Led the Cortex AI team for **Reliance Jio**'s BrainOS: a framework-agnostic learning system (protobuf and ONNX schemas) under Dr. Shailesh Kumar and Akshay Mathur.

Senior Data Scientist, *Gamezop* – Gurgaon, IN (Remote) Jun 2021 – Jun 2022

- Led **ClickVision**: predicted a user's cash deposit within 5 minutes of sign-up; cut wasted offers and churn by about 60%.

Founder & CEO, *Tensr.AI* – Noida, IN (Remote) Feb 2020 – Jun 2021

- Built **Teresa Band**, a cardiac-triaging wearable, with a 15-person team and 3 co-founders; raised 4 government grants; DST DPIIT certified.
- Incubated by SIIC IIT Kanpur (NIDHI EIR), HTIC IIT Madras and Catalyst IIT Mandi.

Data Scientist, *xtLytics LLC* – Noida, IN Jan 2019 – Jan 2020

- Shipped **CADtnet** (edge breast-ultrasound triage app) and **TableOCR** (OCR on grid-less tables).

Publications

[ISBI'26] **TrackletGPT** arXiv:2601.13935 **2026**
[ISBI'26] **TractRLFusion** arXiv:2601.13897

[ISBI'25] **TractoGPT: A GPT architecture for White Matter Tract Segmentation** **2025**
arXiv:2501.15464

• **Anoushkrit Goel**, Simroop Singh, Ankita Joshi, Ranjeet Ranjan Jha, Chirag Ahuja, Aditya Nigam, Arnav Bhavsar. IEEE ISBI 2025.

[ICPR'24] **TractoEmbed: Modular Multi-level Embedding framework for white matter tract segmentation** doi **2024**

• **Anoushkrit Goel**, Bipanjit Singh, Ankita Joshi, Ranjeet Ranjan Jha, Chirag Ahuja, Aditya Nigam, Arnav Bhavsar. ICPR 2024.

[ICPR'24] **Tract-RLFormer: A Tract-Specific RL policy based Decoder-only Transformer Network** doi **2024**

• Ankita Joshi, Ashutosh Sharma, **Anoushkrit Goel**, Ranjeet Ranjan Jha, Chirag Ahuja, Arnav Bhavsar, Aditya Nigam. ICPR 2024.

Work at a Glance

One line per project; details follow in **Project Details**.

- **SlicerTracto** (2024–, IIT Mandi) : open-source 3D Slicer extension for fODF, tracking, segmentation and metrics, built for neurosurgical planning.
- **TrackletGPT** (2025, IIT Mandi): B-spline tokenization of streamlines; ISBI'26.
- **TractRLFusion** (2025, IIT Mandi): GPT-based RL policy fusion for tract-specific tractography; ISBI'26.
- **TractoGPT** (2024, IIT Mandi) : GPT pretrained on streamline point-cloud patches; beats FIESTA; ISBI'25.
- **Tract-RLFormer** (2024, IIT Mandi): supervised plus RL decoder-only transformer for tract-specific tracking; ICPR'24.
- **TractoEmbed** (2024, IIT Mandi) : multi-level embeddings (CNN, dVAE, PointNet); beats Harvard's TractCloud; ICPR'24.
- **TractoAware** (2023–24, IIT Mandi) : per-point normals, gradients and angles with physics-aware loss for tract segmentation.
- **Sculpt3D** (2025, IIT Mandi) : generate and edit point clouds from text, sketch and image.
- **Talk2Point** (2024, IIT Mandi) : finer patch embeddings in ChangeIt3D for text-based point-cloud editing.
- **MedIntel** (2024, IIT Mandi) : med-llama chatbot fine-tuned for differential diagnosis on DDx-plus.
- **VikramUnity** (2023, IIT Mandi) : PPO rover agent navigating a simulated lunar terrain in Unity.
- **Attentive.AI CAD document AI** (2025–): vector-path pretraining (+7% accuracy) and cascaded OCR for architectural drawings.
- **BrainOS Cortex** (2022–23, StatusNeo): framework-agnostic ML schema layer for Jio using protobuf and ONNX.
- **ClickVision** (2021–22, Gamezop) : Markov-chain deposit prediction within 5 minutes of sign-up.
- **Teresa Band** (2020–21, Tensr.AI) : wearable capturing ECG, PPG and seismocardiogram signals.
- **pyCardiograms** (2020, Tensr.AI): peak and wave annotation across ECG and SCG, built on NeuroKit.
- **CVS-valve** (2020, Tensr.AI): 6-compartment Simulink heart model with graded valve malfunction.
- **CADtnet** (2019, xtLytics) : edge TFLite app triaging breast ultrasound by BI-RADS score.
- **TableOCR** (2019, xtLytics): Tesseract and Cloud Vision OCR for grid-less tabular documents.

Grants, Honors and Certifications

- Research funded by SERB / DST India (white matter analysis project); 4 government grants raised for Tensr.AI.
- DST DPIIT Startup India certification (Tensr.AI Technologies Pvt. Ltd.); NIDHI EIR at SIIC IIT Kanpur.
- Rank 3, Himalayan Startup Trek 2019 (Catalyst, IIT Mandi).
- Coursera & Deeplearning.ai Deep Learning Specialization.

Project Details

Grouped by period. Many repositories are private; access on request.

Neuroimaging and 3D Point Clouds (IIT Mandi)

SlicerTracto ongoing Dec 2024 – Present

- Four modules (fODF reconstruction, tracking, segmentation, metrics) with local or SSH-based remote

compute and interchangeable wrapped algorithms.
- Interactive visualization of fiber tracts over brain images; runs on Mac, Windows and Linux.

TractoGPT Jul – Nov 2024
- Streamlines as point clouds, split into patches and tokenized to pretrain a GPT with dual masking; fine-tuned for segmentation on streamline, cluster and fusion representations.
- Outperformed FIESTA (SCIL) on average DICE, overlap and overreach; validated on TractoInferno and 105HCP.

TractoEmbed Apr – Jul 2024
- Hierarchical representation (streamline, cluster, patch) encoded by CNN, dVAE and PointNet; modular so further embeddings can be added.
- Outperformed Harvard Medical School's TractCloud across datasets and age groups.

Tract-RLFormer 2024
- Two-stage policy refinement (supervised then RL) with a decoder-only transformer that delineates tracts directly, skipping segmentation; validated on TractoInferno, HCP and ISMRM2015.

TrackletGPT and TractRLFusion 2025
- TrackletGPT tokenizes streamlines as B-spline tracklets, fusing the TractoGPT work with a new sequence representation.
- TractRLFusion fuses tract-specific RL policies with a GPT backbone; both appear at ISBI'26.

TractoAware Oct 2023 – Feb 2024
- Extended (x,y,z) points to 12 features (neighbours, normals, angles); transformer and varying-kernel PointNet embeddings; physics-aware loss alongside cross-entropy.
- Reported about a 15% jump in classification and segmentation accuracy; BUAN scores computed against saved reference tracts.

Multimodal, Language and Simulation (IIT Mandi)

Sculpt3D 2025
- Generation and continuous editing of point clouds from sketch, image, text and point-cloud input, built on SDFusion and ChangeIt3D with a finer-grained encoder.

Talk2Point Feb – May 2024
- Modified ChangeIt3D tokenization on ShapeTalk for region-specific edits; SDFusion for reconstruction from scratch.

MedIntel Feb – May 2024
- Fine-tuned med-llama and LLaVA-Med on DDx-plus for long-form differential diagnosis across common ailments; served via ollama, deployed on PARAM Himalaya, Streamlit UI.

VikramUnity Sep – Nov 2023
- Chandrayaan-3 Vikram rover on a custom lunar terrain (Unity, Blender, ml-agents); PPO agent rewarded for daylight travel, few bumps and falls, and reaching mining targets. CS662 course project.

Industry

CAD and Architectural Document AI (Attentive.AI) 2025 – Present
- Pretraining on tokenized vector paths (+7% class-wise accuracy), symbol search, HVAC annotation, and cascaded multi-OCR callout extraction.

BrainOS Cortex (StatusNeo for Reliance Jio) 2022 – 2023
- Library-agnostic learning framework: algorithms defined as protobuf and ONNX schemas, independent of scikit-learn, PyTorch or TensorFlow.

ClickVision (Gamezop) Jun 2021 – Jun 2022
- MDP-inspired transition matrices on clickstream data; trajectories are scored by weight, discount factor and transition probability, with decay on other paths.
- Cut wasted offers and churn by about 60% (tracked in MoEngage); verdict within 2 minutes of use; deployed internally on Gamezop.

Entrepreneurship and Earlier Work

Teresa Band, pyCardiograms and CVS-valve (Tensr.AI) 2020 – 2021

- Teresa Band: flexible TR-90 band with 12-bit ADC and microprocessor capturing ECG, PPG and seismocardiograms, wired to an Android phone.
- pyCardiograms: wave annotation and inter-wave correlation across ECG and SCG on PhysioBank; Bi-LSTM prediction of two heartbeat cycles of SCG.
- CVS-valve: 6-compartment lumped-parameter Simulink model with graded valve aperture to simulate regurgitation and stenosis and relate hemodynamics to SCG.

CADtnet and TableOCR (xtLytics) 2019

- CADtnet: phone photos of breast ultrasound classified by BI-RADS score with lesion bounding box; TFLite and MobileNet v2 Android app (CADx) for the Susan Love Foundation, tested in Guadalajara, Mexico.
- TableOCR: Tesseract and Cloud Vision output with geometric bounding-box matching to rebuild rows; limited on multi-column rows.

Ongoing and Planned

- **Med3D**: 3D point-cloud input for medical LLMs via Point-LLM. **TractoSAM**: SAM for 3D tract segmentation. **BiometricSEAL**: embedding biometrics in generated images. **TumorGen** and **TumorProg**: tumor generalization and progression modelling.

Skills

Languages: Python, MATLAB, C, C#, ProtoBuf, LaTeX
ML and data: PyTorch, TensorFlow/Keras, ONNX, scikit-learn, scipy, pandas, PySpark, Open3D, PyTorch3D, OpenCV, Neurokit, wfdb, Unity ML-Agents, LangChain, LangGraph, ollama
Methods: point-cloud transformers, GPT pretraining, deep RL (TD3, PPO, SAC), RAG, fine-tuning LLMs, multimodal tokenization
Neuroimaging and tools: 3D Slicer, dipy, scilpy, MRtrix3, FSL, FreeSurfer, Simulink, AWS, GCP, Verilog
Hobbies: portrait sketching, wood sculpting, oil painting, gym

References

- **Arnav Bhavsar**, Professor, IIT Mandi
- **Aditya Nigam**, Professor, IIT Mandi
- **Ranjeet Ranjan Jha**, Assistant Professor, IIT Patna
- **Shailesh Kumar**, Chief Data Scientist, Jio